Artificial Intelligence — Lab 1 Report

GitHub-ready submission companion

1. Problem Statement

The lab covers AI historical milestones, transparent rule-based scholarship evaluation, and the distinction between conversational fluency and general intelligence.

2. Assumptions and Implementation

The rule uses score >= 70, complete documentation, attendance >= 80%, and recommendation >= 3/5. The last two are the added named conditions. All thresholds are synthetic lab assumptions.

3. Timeline

1956 Dartmouth AI project; 1966 ELIZA; 1974–1980 First AI Winter; 1987–1993 Second AI Winter; 2012 AlexNet/ImageNet; 2017 Transformer; 2020s large-scale compute/data era. Sources are listed in the DOCX report.

4. Four Core Tests

1 Pass: 85/complete/90%/4 -> PASS. 2 Attendance failure: 85/complete/79%/4 -> FAIL. 3 Recommendation failure: 85/complete/90%/2 -> FAIL. 4 Boundary: 70/complete/80%/3 -> PASS.

5. Disagreement and Comparison

A fluent conversational response can miss an explicit constraint. For score=70, complete documents, attendance=80%, recommendation=2/5, the conversational-style response suggests qualification, while the rule evaluator correctly returns FAIL.

6. Failure / Edge Case

Likely cause: salient facts are handled while one constraint is omitted. Improvement: explicit machine-checkable predicates plus a validation step and auditable reason list.

7. Reflection and Responsible Practice

Use reproducible tests, edge cases, documented assumptions, privacy protection, explanations, human oversight, and review of real-world fairness before deploying an automated scholarship decision system.

References

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